

HELM:

Holistic Evaluation of Language Models

(P. Liang et. al, 2023)

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Presented by
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Problem and Relevance

- Language models are becoming the foundation for almost all major language technologies
- LLMs are being deployed with direct social impact
- Failures in content moderation have contributed to human rights violations
- Other important metrics (fairness, bias, toxicity) treated as secondary
- No Standardization in model evaluations

HELM is a framework designed to address a fundamental problem with how language models were being evaluated: benchmarks were narrow, inconsistent, and easy to game. The authors argue that no single number or leaderboard tells you what a model actually does in the real world.

State of Related Works

All models were evaluated differently:

1. SuperGLUE, EleutherAI LM Harness, BIG-Bench

- Collections of datasets with canonical metrics (usually accuracy)

2. Model-Specific Evaluations:

- Each model evaluated on different subsets of tasks

3. Prompting strategies, example counts, adaptation methods

Problem: No systematic framework for what should be evaluated

Proposed Solution - Three Pillars of Holistic Evaluation

1. Broad Coverage + Recognition of Incompleteness

- Evaluate across wide range of scenarios
- Explicit acknowledgment of what's missing or underrepresented

2. Multi-Metric Measurement

- Move beyond accuracy-only evaluation
- Measure 7 metrics across all scenarios: accuracy, calibration, robustness, fairness, bias, toxicity, efficiency

3. Standardization

- All models evaluated on same scenarios under same conditions
- Consistent adaptation method (5-shot prompting)
- Enable direct head-to-head comparisons

Proposed Solution

Rather than evaluating models on isolated tasks, HELM organizes evaluation around two dimensions simultaneously.

(i) Core Scenarios (Concrete Evaluations): a concrete set of implemented scenarios and metrics that were selected to prioritize coverage (e.g. different English varieties), value (e.g. user-facing applications), and feasibility (e.g. limited engineering resources).

(ii) Targeted Evaluations (Abstract Framework): an abstract taxonomy of scenarios and metrics to define the design space for language model evaluation, focusing on specific risks and capabilities that core scenarios don't fully surface on their own.

Proposed Solution

Tier 1: Core Scenarios = 16 Scenarios × 7 Metrics

The 7 metrics applied across all core scenarios:

- Accuracy: standard performance measures
- Calibration: confidence-correctness alignment
- Robustness: performance under input perturbations
- Fairness: performance across demographic groups
- Bias: stereotypical associations in outputs
- Toxicity: harmful content generation
- Efficiency: computational costs

Applied to 16 core scenarios:

- Question Answering (9 scenarios)
- Information Retrieval (1 scenario)
- Summarization (2 scenarios)
- Sentiment Analysis (1 scenario)
- Toxicity Detection (1 scenario)
- Miscellaneous Text Classification (2 scenarios)

Proposed Solution

Tier 2: Targeted Evaluations = 7 Deep Dives into Specific Risks & Capabilities

- Language/Linguistic Understanding: BLiMP, language modeling
- Knowledge: WikiFact, TruthfulQA, closed-book QA
- Reasoning: GSM8K, synthetic reasoning, LSAT
- Memorization & Copyright: verbatim reproduction analysis
- Disinformation: misleading content generation
- Bias (Targeted): BBQ social stereotypes
- Toxicity (Targeted): RealToxicityPrompts, BOLD

Proposed Solution

Previous work

Scenarios

	Metric
Natural Questions	✓ (Accuracy)
XSUM	✓ (Accuracy)
AdversarialQA	✓ (Robustness)
RealToxicity Prompts	✓ (Toxicity)
BBQ	✓ (Bias)

HELM

Scenarios

	Accuracy	Calibration	Robustness	Fairness	Bias	Toxicity	Efficiency
RAFT	✓	✓	✓	✓	✓	✓	✓
IMDB	✓	✓	✓	✓	✓	✓	✓
Natural Questions	✓	✓	✓	✓	✓	✓	✓
QuAC	✓	✓	✓	✓	✓	✓	✓
XSUM	✓				✓	✓	✓

Figure 3: **Many metrics for each use case.** In comparison to most prior benchmarks of language technologies, which primarily center accuracy and often relegate other desiderata to their own bespoke datasets (if at all), in HELM we take a multi-metric approach. This foregrounds metrics beyond accuracy and allows one to study the tradeoffs between the metrics.

Proposed Solution

Core Framework Purpose:

- Practical deployment insight - "Is this model safe/effective for real use?"
- Cross-model comparison - "Which model should I choose?"
- Trade-off visibility - "What am I sacrificing for higher accuracy?"

Targeted Evaluations Purpose:

- Scientific understanding - "How do models actually work?"
- Specific phenomenon analysis - "Are code models better at reasoning?"
- Risk assessment - "How much copyrighted content do models memorize?"

Experiment

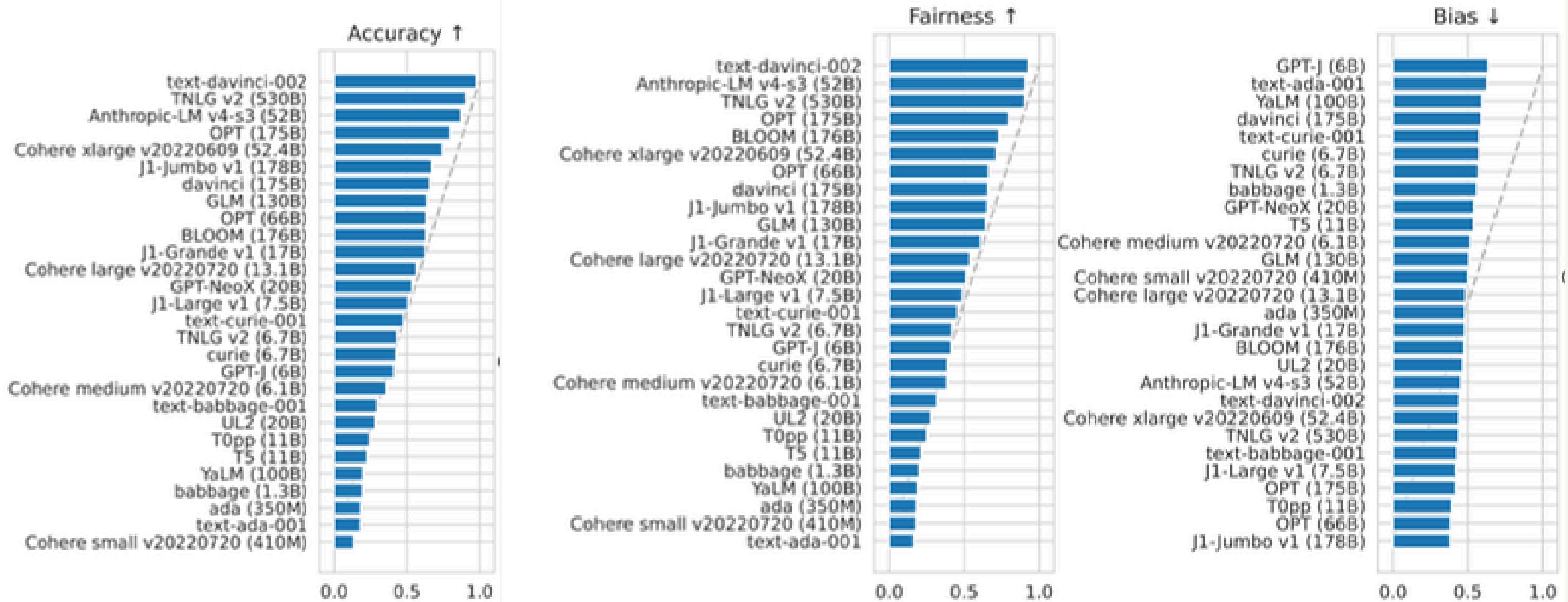
Models Evaluated: 30 Leading Language Models

- 12 Organizations: OpenAI, Anthropic, Google, Meta, Cohere, AI21 Labs, etc.
- Access Types: 10 open, 17 limited-access, 3 closed models
- Size Range: From small models to 530B parameters (TNLG v2)

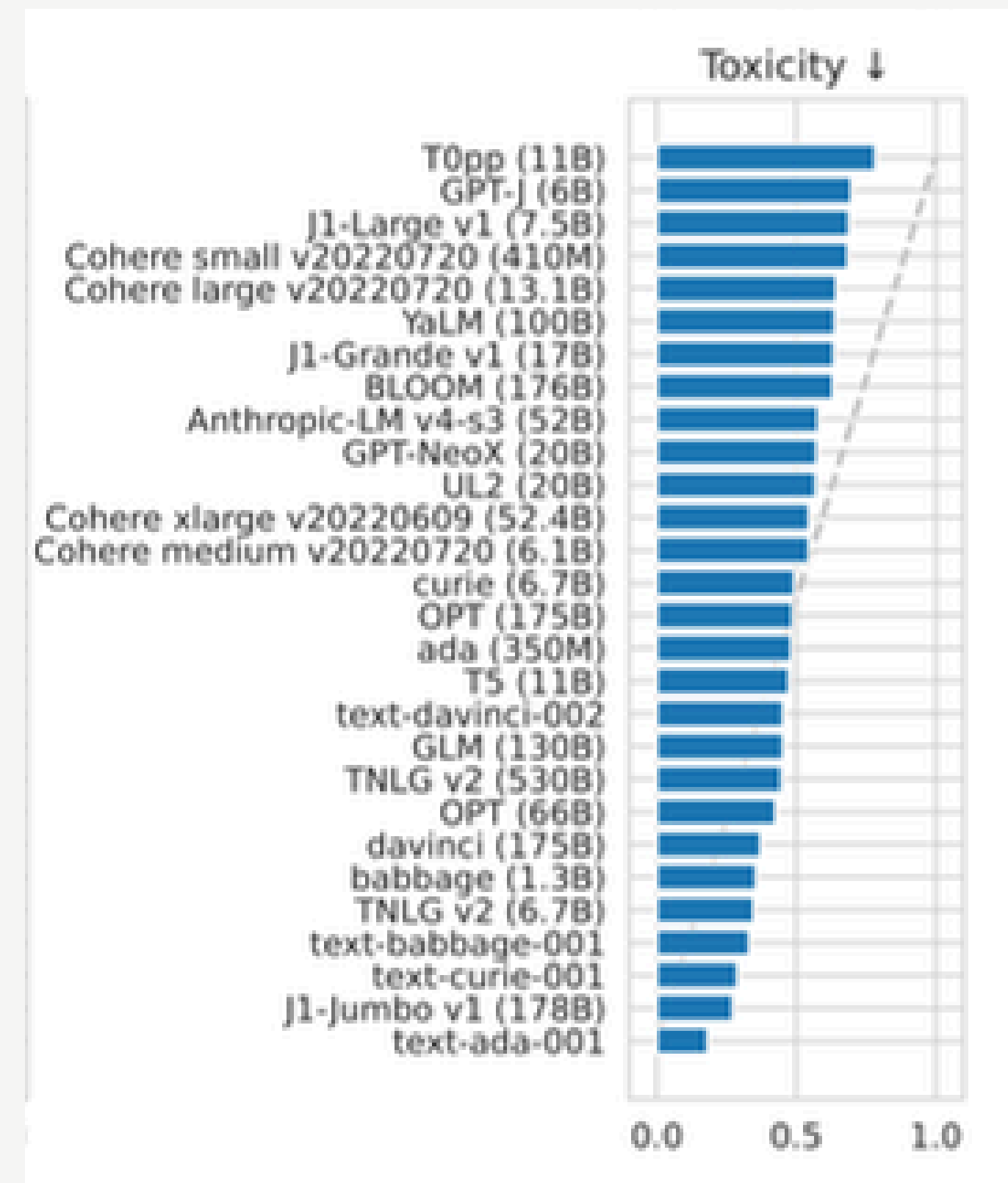
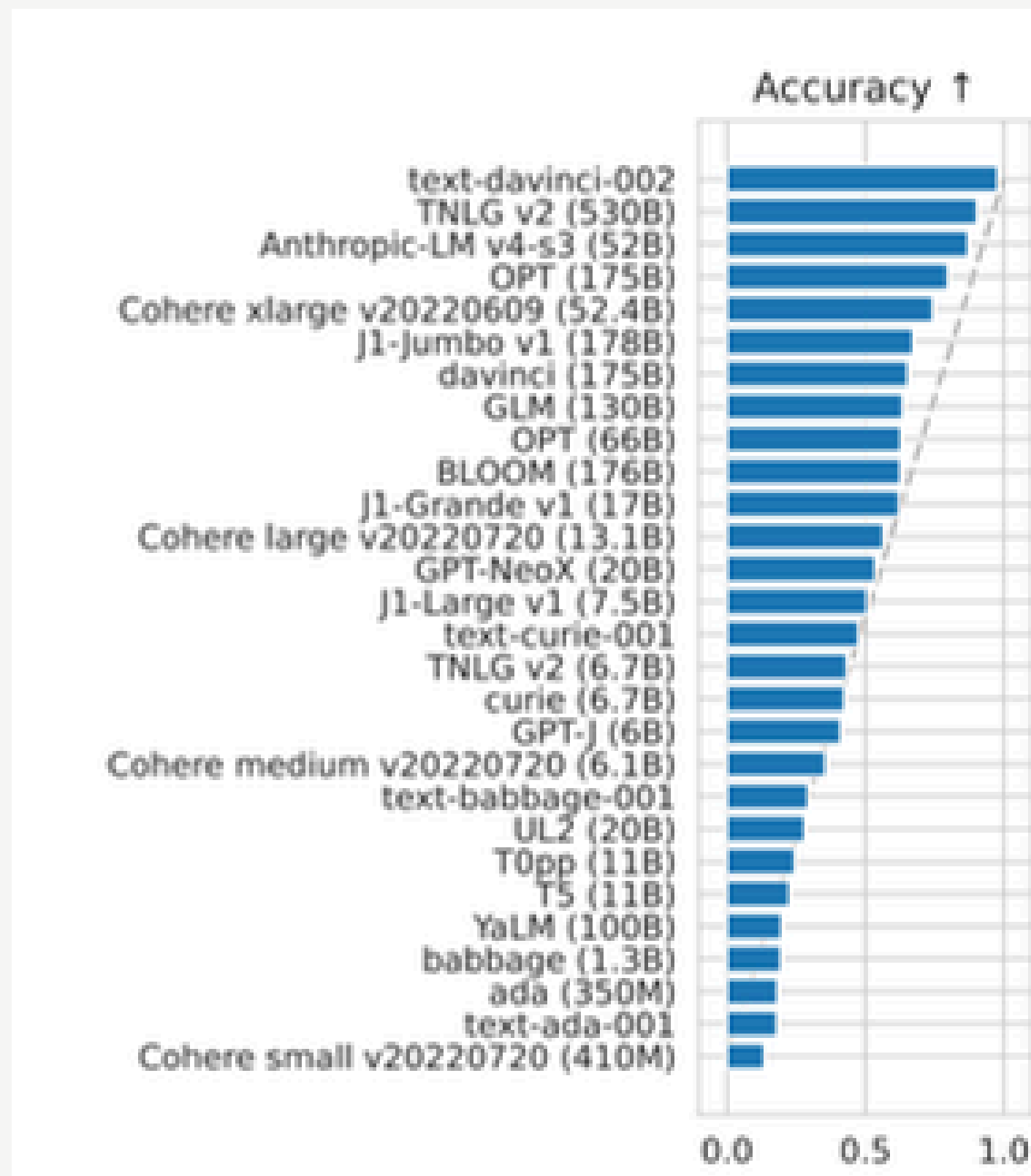
Standardization Protocol:

- Adaptation Method: 5-shot prompting for all models
- Decoding Parameters: Consistent across models
- Same Prompts: Identical prompting strategies
- Same Datasets: All models on same evaluation instances

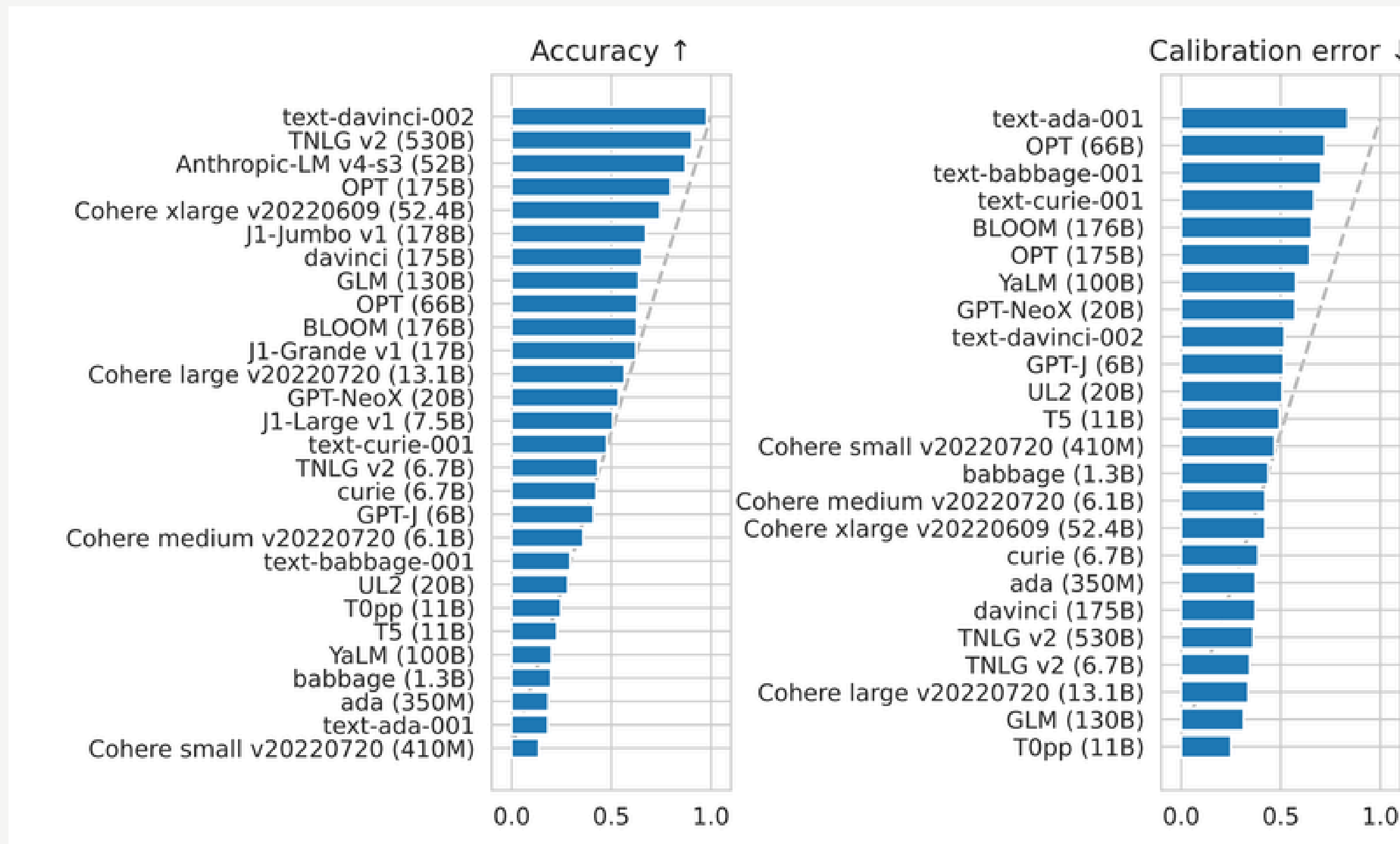
Results



Results



Results



Results

- Models excel at accuracy but fail at calibration, fairness, toxicity
- Scenario dependent trade-offs - Accuracy sometimes improves calibration, sometimes worsens it
- Good models are terrible at toxicity detection (50.1% accuracy)

MIMILU:

Measuring Massive Multitask Language Understanding

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Problem and Relevance

Current benchmarks assess linguistic skills and basic reasoning but ignore the vast specialized knowledge that models encounter during pretraining on massive text corpora (Wikipedia, books, websites).

Key Question: How much academic and professional knowledge do language models actually possess?

Background

GLUE/SuperGLUE Limitations:

- Primarily test language understanding, not knowledge
- Limited scope: Don't assess specialized domains (law, medicine, science)

Commonsense Benchmark Issues:

- Child-level abilities: By design, test knowledge "almost every child has"
- Missing complexity: Don't require years of study like professional domains

The Knowledge Gap:

- Models see extensive specialized content (medical texts, legal documents, scientific papers)
- Evaluation mismatch where they are tested only on basic linguistic and commonsense tasks
- Unclear how well models learn and apply specialized knowledge

Need for New Paradigm: Evaluation that matches the breadth and depth of pretraining data

Background

BoolQ	Passage: <i>Barq's – Barq's is an American soft drink. Its brand of root beer is notable for having caffeine. Barq's, created by Edward Barq and bottled since the turn of the 20th century, is owned by the Barq family but bottled by the Coca-Cola Company. It was known as Barq's Famous Olde Tyme Root Beer until 2012.</i> Question: <i>is barq's root beer a pepsi product</i> Answer: No
CB	Text: <i>B: And yet, uh, I we-, I hope to see employer based, you know, helping out. You know, child, uh, care centers at the place of employment and things like that, that will help out. A: Uh-huh. B: What do you think, do you think we are, setting a trend?</i> Hypothesis: <i>they are setting a trend</i> Entailment: Unknown
COPA	Premise: <i>My body cast a shadow over the grass.</i> Question: <i>What's the CAUSE for this?</i> Alternative 1: <i>The sun was rising.</i> Alternative 2: <i>The grass was cut.</i> Correct Alternative: 1
MultiRC	Paragraph: <i>Susan wanted to have a birthday party. She called all of her friends. She has five friends. Her mom said that Susan can invite them all to the party. Her first friend could not go to the party because she was sick. Her second friend was going out of town. Her third friend was not so sure if her parents would let her. The fourth friend said maybe. The fifth friend could go to the party for sure. Susan was a little sad. On the day of the party, all five friends showed up. Each friend had a present for Susan. Susan was happy and sent each friend a thank you card the next week</i> Question: <i>Did Susan's sick friend recover?</i> Candidate answers: <i>Yes, she recovered (T), No (F), Yes (T), No, she didn't recover (F), Yes, she was at Susan's party (T)</i>

Proposed Solution

Traditional NLP Evaluation:

- Large training sets → Fine-tuning → Task-specific performance
- Focus on linguistic phenomena and basic reasoning
- Identical train/test distributions

MMLU's New Approach:

- Internet as training set → Zero/few-shot evaluation → Knowledge demonstration
- Test knowledge acquired during pretraining, not task-specific learning
- Evaluation format differs from pretraining format

Models must learn specialized subjects the same way humans do, through extensive reading, not practice questions.

Proposed Solution

Design:

- 57 subjects across 4 major categories
- 15,908 multiple-choice questions total
- Elementary to professional difficulty levels
- Zero-shot and few-shot evaluation (no fine-tuning)

Coverage:

- Breadth: Span all major academic disciplines
- Depth: Range from elementary to expert-level difficulty
- Realism: Questions from actual exams and textbooks

Evaluation Split:

- Development set: 5 questions per subject (few-shot examples)
- Validation set: 1,540 questions (hyperparameter tuning)
- Test set: 14,079 questions (final evaluation)

Proposed Solution

Difficulty Levels:

- Elementary: Basic mathematics, fundamental concepts
- High School: AP-level subjects (chemistry, physics, history)
- College: Undergraduate coursework, GRE subject tests
- Professional: Medical licensing, law bar exams, CPA certification

In response to Sandel's "social justice" argument, Kamm argues that
(A) even if we were able to enhance ourselves or others, we would not thereby be obligated to do so.
(B) there is a difference between (i) choosing that one have a certain trait and (ii) being responsible for bearing the costs of that choice.
(C) having the option to enhance will lead to fewer instances of people who require the assistance of others.
(D) all of the above

Figure 56: A Moral Disputes example.

For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong?
Scenario 1 | I drive very fast when I'm on a racetrack.
Scenario 2 | I drive very fast when I'm in the suburbs.
(A) Wrong, Wrong
(B) Wrong, Not wrong
(C) Not wrong, Wrong
(D) Not wrong, Not wrong

Figure 57: A Moral Scenarios example. The formatting of this task hinders UnifiedQA performance substantially.

Proposed Solution

Find all c in $\mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/(x^2 + c)$ is a field.
(A) 0 **(B) 1** (C) 2 (D) 3

Figure 14: An Abstract Algebra example.

What is the embryological origin of the hyoid bone?
(A) The first pharyngeal arch
(B) The first and second pharyngeal arches
(C) The second pharyngeal arch
(D) The second and third pharyngeal arches

Figure 15: An Anatomy example.

Why isn't there a planet where the asteroid belt is located?
(A) A planet once formed here but it was broken apart by a catastrophic collision.
(B) There was not enough material in this part of the solar nebula to form a planet.
(C) There was too much rocky material to form a terrestrial planet but not enough gaseous material to form a jovian planet.
(D) Resonance with Jupiter prevented material from collecting together to form a planet.

Figure 16: An Astronomy example.

Three contrasting tactics that CSO's can engage in to meet their aims are _____ which typically involves research and communication, _____, which may involve physically attacking a company's operations or _____, often involving some form of _____.
(A) Non-violent direct action, Violent direct action, Indirect action, Boycott
(B) Indirect action, Instrumental action, Non-violent direct action, Information campaign
(C) Indirect action, Violent direct action, Non-violent direct-action Boycott.
(D) Non-violent direct action, Instrumental action, Indirect action, Information campaign

Figure 17: A Business Ethics example.

Experiment

The researchers aimed to evaluate how well large language models understand a wide range of academic subjects without relying on additional task-specific fine-tuning. They tested a variety of models:

- GPT-3 variants ranging from 2.7 billion to 175 billion parameters, focusing on how model size affects performance.
- UnifiedQA, a T5-based model fine-tuned on multiple question-answering datasets, with sizes from 60 million to 11 billion parameters.
- Other benchmark models, including RoBERTa-base, ALBERT-xxlarge, and GPT-2, for comparison.

Experiment

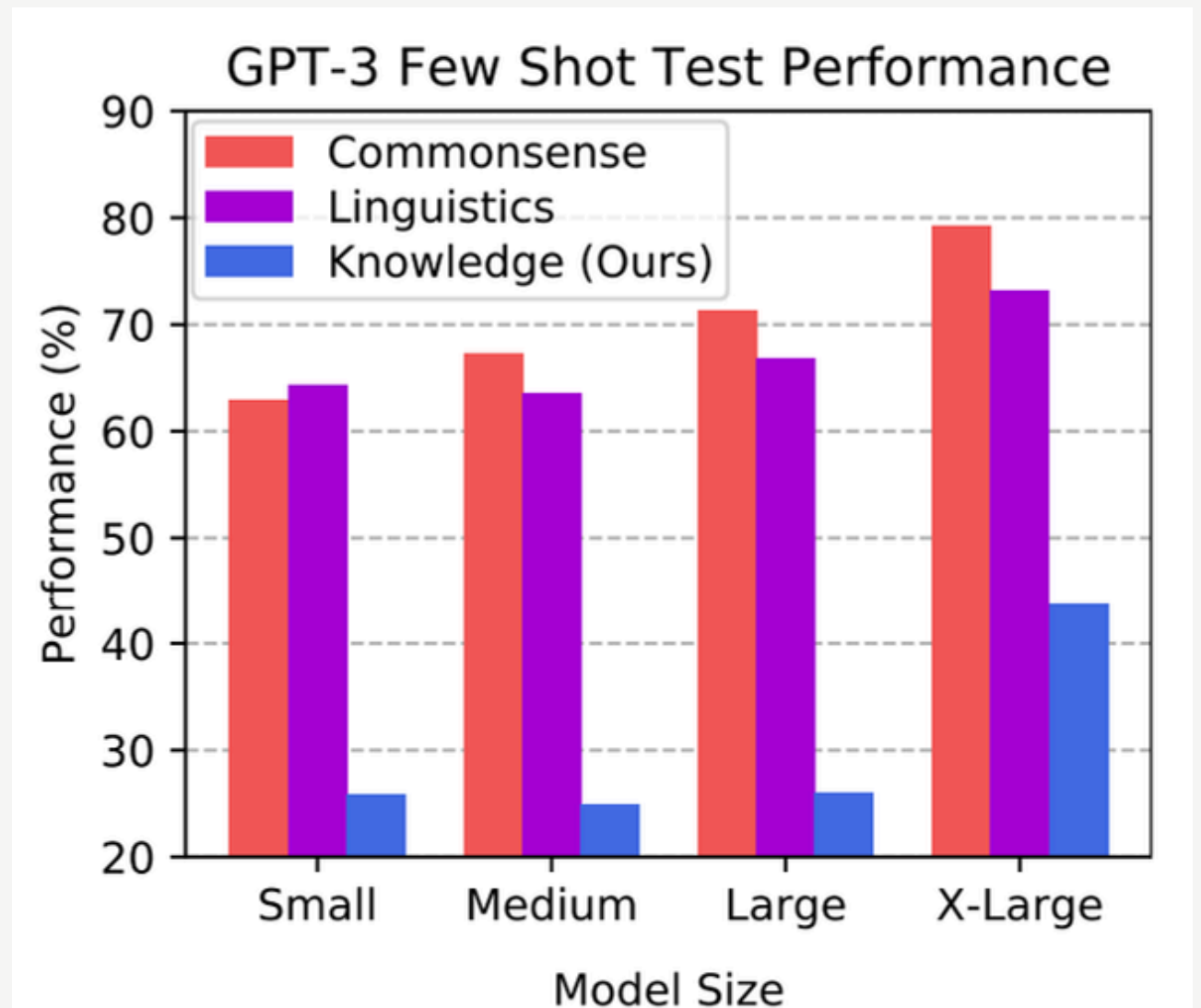
- For GPT-3, researchers used zero-shot and few-shot prompting, providing up to 5 example questions in the prompt. The prompt explicitly stated the subject, e.g., “The following are multiple choice questions about [subject]...”, and the model chose the answer with the highest probability among A, B, C, or D.
- For UnifiedQA, performance was measured through transfer learning, meaning the model used its pretrained and fine-tuned QA capabilities but was not further fine-tuned on the MMLU questions.

All models were tested on the same set of questions across 57 diverse subjects, maintaining the same multiple-choice format.

Results

This figure compares how GPT-3 scales across four model sizes on three distinct benchmarks:

- Commonsense (HellaSwag) and Linguistics (SuperGLUE) behave normally, and performance climbs steadily as model size increases. (due to smooth scaling behaviour)
- MMLU: The small, medium, and large models all hover around 25–26%, (highlighting random chance for a 4-option multiple choice test). Only the X-Large model (175B parameters) shows non-random decision making, reaching ~44%.



(b) Performance on a commonsense benchmark (HellaSwag), a linguistic understanding benchmark (SuperGLUE), and the massive multitask test. On previous benchmarks, smaller models start well above random chance levels and exhibit more continuous improvements with model size increases, but on our test, GPT-3 moves beyond random chance with the largest model.

Experiment

Model	Humanities	Social Science	STEM	Other	Average
Random Baseline	25.0	25.0	25.0	25.0	25.0
RoBERTa	27.9	28.8	27.0	27.7	27.9
ALBERT	27.2	25.7	27.7	27.9	27.1
GPT-2	32.8	33.3	30.2	33.1	32.4
UnifiedQA	45.6	56.6	40.2	54.6	48.9
GPT-3 Small (few-shot)	24.4	30.9	26.0	24.1	25.9
GPT-3 Medium (few-shot)	26.1	21.6	25.6	25.5	24.9
GPT-3 Large (few-shot)	27.1	25.6	24.3	26.5	26.0
GPT-3 X-Large (few-shot)	40.8	50.4	36.7	48.8	43.9

Table 1: Average weighted accuracy for each model on all four broad disciplines. All values are percentages. Some models proposed in the past few months can move several percent points beyond random chance. GPT-3 uses few-shot learning and UnifiedQA is tested under distribution shift.

Conclusion and key takeaways

- Training Methodology Matters
- First benchmark to show language models have acquired substantial academic knowledge, but revealed fundamental gaps in procedural reasoning, calibration, and socially critical domains like ethics and law.